Hao-Tian (Jim) Xiao

Machine Learning Engineer
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SUMMARY

Machine Learning Engineer at QCT, building production agentic AI systems and on-prem LLM/VLM serving.

M.S. research focused on machine learning and computer vision, including model fine-tuning methods and knowledge distillation, applied hands-on to smart aquaculture through cross-disciplinary collaboration, from understanding real needs to designing e2e solutions.

EXPERIENCE

Machine Learning Engineer — Quanta Cloud Technology (QCT), Taoyuan Sep 2024 – Present

- Build agentic systems on NeMo Agent Toolkit for software-engineering workflows (research, testing, model fine-tuning); one production deployment serves **~100 internal users**, and an on-prem agent automates **MLPerf** benchmarking on Quanta/QCT GPU servers.
- Extend SOTA models for semantic/instance segmentation and image/video understanding; design the pipeline on NVIDIA VSS and apply knowledge distillation for real-time inference.
- Own inference tuning for production LLM/VLM services; standardized on vLLM after evaluating NVIDIA NIM, TensorRT-LLM, Ollama, and llama.cpp; optimize TTFT, throughput, and e2e latency under real concurrent workloads.

Research Intern, 5G + AI — Quanta Cloud Technology (QCT), Taoyuan Jun – Nov 2023

- Deployed free5GC and built NWDAF to collect UE/network metrics; simulated multi-UE behavior across network slices with UERANSIM.
- Applied deep-learning time-series models to predict upcoming network events for **better QoS**.
- **Honorable Mention**, Institute for Information Industry (資策會) 5G Jump Competition.

Graduate Research Assistant — National Taiwan Ocean University · Intelligent System Lab 2022 – 2024

- Led a cross-disciplinary team turning computer-vision research into field deployment for smart aquaculture.

  Field-interviewed farmers and coordinated departments and vendors.

Teaching Assistant — National Taiwan Ocean University · CSE 2019 – 2024

- Assisted Programming (C/C++), Machine Learning, Machine Vision, and General Physics.
- Built course materials and exams and ran after-class tutoring.

PROJECTS

Smart Fish Feeding System — Fisheries Agency R&D · National Taiwan Ocean University 2022 – 2024

- Delivered two Fisheries-Agency-funded R&D projects: a cloud-based computer-vision feeding system deployed across **7 land-based farms** in southern Taiwan, using knowledge distillation and feeding-splash segmentation; exhibited at the 2023 Taiwan Innotech Expo (TIE).

Patent · TWI892930 — Smart Fish Feeding Intensity Identification System and Method Granted

EDUCATION

M.S., Computer Science & Engineering — National Taiwan Ocean University 2022 – 2024

- Thesis: A Cloud-based Smart Fish Feeding System using Knowledge Distillation and Feeding-Splash Segmentation (presented at IPPR CVGIP 2024).

B.S., Computer Science & Engineering — National Taiwan Ocean University 2018 – 2022

SKILLS

ML / AI: Agentic systems (NeMo Agent Toolkit), LLM/VLM serving & fine-tuning, semantic/instance segmentation, image/video understanding, knowledge distillation

Serving & Infra: vLLM, NVIDIA NIM, TensorRT-LLM, Ollama, llama.cpp, on-prem GPU, Docker, Kubernetes, AIPerf, MLPerf

Languages & Tools: Python, C/C++, PyTorch, Git

5G Research: 5G core (free5GC), NWDAF, UERANSIM, network slicing, time-series prediction